STA 431 Midterm Notesheet

SIDE A: counting and probability

Homework 1–2; Freund Chapters 1–2. LaTeX reference for a handwritten, double-sided sheet.

1. Counting: choose the operation

AND (successive choices): multiply. **OR** (disjoint cases): add. Count distinct outcomes once; divide if a construction counts each outcome repeatedly. For equally likely elementary outcomes, $P(A) = \#A/\#S$.

Ordered, no replacement	${}_nP_r = n!/(n-r)!$
Unordered, no replacement	$\binom{n}{r} = n!/[r!(n-r)!]$
Ordered, with replacement	n^r
All n distinct in a line	$n!$; around a circle $(n-1)!$
Repeated letters	$n!/\prod_i n_i!$
Labeled groups	$n!/\prod_i n_i!$

Here $0! = 1$ and $\sum_i n_i = n$. For *unlabeled* equal-size groups, divide by permutations of interchangeable groups. Keep groups labeled when roles/sizes distinguish them.

Constrained counts. "Exactly" fixes counts; choose positions. "At least" splits into possible count cases, or use total minus the excluded cases. For adjacent identical letters, tie them into one block, then account for any *other* repeats. For a two-choice test, a wrong answer is forced.

HW: PROBABILITY: $11!/(2!2!) = 9{,}979{,}200$; with the B's adjacent, treat BB as one item: $10!/2! = 1{,}814{,}400$.

HW: Committees of sizes $3, 4, 5$: $12!/(3!4!5!) = 27{,}720$.

HW: 20 T/F, 15 right, exactly 2 of 5 hard right: $\binom{5}{2}\binom{15}{13} = 1050$.

HW: 5 chosen from 8 math, 6 stats; at least 2 of each: $\binom{8}{2}\binom{6}{3} + \binom{8}{3}\binom{6}{2} = 1400$.

HW: 10 students, exactly 3 A and 1 F; each remaining student gets B/C/D: $\binom{10}{3}\binom{7}{1}3^6 = 612{,}360$.

2. Binomial and multinomial tools

$(a+b)^n = \sum_{r=0}^{n}\binom{n}{r}a^{n-r}b^r$; $\binom{n}{r} = \binom{n}{n-r}$ and $\sum_r \binom{n}{r} = 2^n$.

Pascal's identity. For $1 \le r \le n-1$, $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$. Story: fix one item; an r-subset either includes it or does not. Algebra: put both fractions over $r!(n-r)!$; numerator is $(n-1)![r + (n-r)] = n!$.

Proving sums from $(1+x)^n$. Differentiate, then $x = 1$: $\sum_{r=1}^{n} r\binom{n}{r} = n2^{n-1}$. Integrate from 0 to 1: $\sum_{r=0}^{n}\binom{n}{r}/(r+1) = (2^{n+1}-1)/(n+1)$.

Multinomial term. In $(a_1x_1 + \cdots + a_kx_k)^n$, the $x_1^{r_1}\cdots x_k^{r_k}$ coefficient is $\dfrac{n!}{\prod r_i!}\prod a_i^{r_i}$, if $\sum r_i = n$. Keep signs and constants.

HW: $[x^3y^2z^2](2x - y + 3z)^7 = \dfrac{7!}{3!2!2!}2^3(-1)^2 3^2 = 15{,}120$.

3. Events and set translation

S: all elementary outcomes; A^c: not A; $A \cap B$: both; $A \cup B$: at least one. $A \setminus B = A \cap B^c$. **De Morgan:** $(A \cup B)^c = A^c \cap B^c$; $(A \cap B)^c = A^c \cup B^c$.

For three events: "only A" $= A \cap B^c \cap C^c$; "exactly one" $= (A \cap B^c \cap C^c) \cup (A^c \cap B \cap C^c) \cup (A^c \cap B^c \cap C)$; "at least one" $= A \cup B \cup C$; "A and B but not C" $= A \cap B \cap C^c$; "none" $= (A \cup B \cup C)^c$.

Add and subtract overlaps. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$. Pairwise "both" counts *include* the triple intersection.

Only one region. $P(A \text{ only}) = P(A) - P(A \cap B) - P(A \cap C) + P(A \cap B \cap C)$. **HW:** Weekend only $= 600 - 250 - 150 + 100 = 300$; union $= 1000$, so none $= 0$ (of 1000).

Exactly / at least two of three. Write $T = P(A \cap B \cap C)$ and $J = P(A \cap B) + P(A \cap C) + P(B \cap C)$. Then $P(\text{exactly two}) = J - 3T$, while $P(\text{at least two}) = J - 2T$.

4. Probability rules

$P(S) = 1$, $P(\varnothing) = 0$, $0 \le P(A) \le 1$, $P(A^c) = 1 - P(A)$. If $A \subseteq B$, then $P(A) \le P(B)$. For disjoint A, B, $P(A \cup B) = P(A) + P(B)$.

Conditioning changes the universe. $P(A \mid B) = P(A \cap B)/P(B)$, $P(B) > 0$. Hence $P(A \cap B) = P(B)P(A \mid B) = P(A)P(B \mid A)$. If sampling without replacement, update what remains after each draw.

Contingency table. Put counts in disjoint cells, then divide by the *condition's* row/column total. **HW:** 120 people: graduate F/M $= 40/30$; faculty F/M $= 25/25$. Thus $P(\text{male faculty}) = 25/120 = 5/24$, $P(\text{grad} \mid \text{female}) = 40/65 = 8/13$.

HW: If $P(X) = .55$, $P(Y) = .35$, $P(X \cup Y) = .70$, then $P(X \cap Y) = .20$; $P(X \mid Y) = .20/.35 = 4/7$; $P(Y \mid X^c) = (.35 - .20)/(.45) = 1/3$.

5. Independence is different from disjointness

Independent iff $P(A \cap B) = P(A)P(B)$; then, when defined, $P(A \mid B) = P(A)$. Disjoint means $P(A \cap B) = 0$. Positive-probability disjoint events cannot be independent. To test independence, compare intersection with product; do not infer it from a diagram.

6. Total probability and Bayes

For a partition $B_1, \ldots, B_k$ (disjoint, exhaustive), branch weight $w_i = P(B_i)P(A \mid B_i)$. Then

$$P(A) = \sum_i w_i, \qquad P(B_j \mid A) = \frac{w_j}{\sum_i w_i}.$$

Procedure: draw prior branches B_i (priors sum to 1); attach conditional likelihood of A; multiply along each path; add all A paths for denominator. Posterior = target path / all A paths.

Two-branch template. For B vs B^c, let $p = P(B)$, $s = P(A \mid B)$, $q = P(A \mid B^c)$. Then $P(A) = ps + (1-p)q$ and $P(B \mid A) = ps/[ps + (1-p)q]$. This is the base-rate calculation.

HW: Journal: acceptance $= .50(.15) + .30(.25) + .20(.10) = .17$; $P(\text{methodology} \mid \text{accepted}) = .075/.17 = 15/34$.

HW: Die selects urn A with 1/3, B with 2/3; red rates $3/7, 5/7$. $P(R) = \frac{1}{3}\frac{3}{7} + \frac{2}{3}\frac{5}{7} = 13/21$; $P(A \mid R) = (1/7)/(13/21) = 3/13$.

HW: Spam prior .60; "urgent" rates .40 in spam, .05 otherwise. $P(U) = .60(.40) + .40(.05) = .26$; $P(\text{spam} \mid U) = .24/.26 = 12/13$.

Three-way independence. For A, B, C to be mutually independent, all three pair products *and* $P(A \cap B \cap C) = P(A)P(B)P(C)$ must hold. Pairwise independence alone does not ensure the triple equality.

7. Fast setup checks

Count: order? replacement? distinguishable? are cases disjoint? any overcount?
Probability: write the conditioning event in the denominator; check $0 \le P \le 1$.
Three sets: add singles, subtract pairs, add triple; a triple belongs to every pair count.
Trees: multiply along branches and add across branches.
"At least one": often use $1 - P(\text{none})$.
Independent components: series success $= \prod p_i$; parallel success $= 1 - \prod(1 - p_i)$, only when components are independent.

Source: Freund, 8th ed., Chs. 1–2. HW examples are from STA 431 required problems only.

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Homework 3; Freund Chapter 3. For any CDF, $F(x) = P(X \le x)$ includes the endpoint.

8. Discrete random variables

A random variable assigns a number to each outcome. List its possible values (*support*) and group all elementary outcomes producing the same value. PMF $p_X(x) = P(X = x)$: require $p_X(x) \ge 0$ and $\sum_x p_X(x) = 1$. To find $P(X \in A)$, sum the masses for $x \in A$.

Discrete CDF. $F_X(t) = P(X \le t) = \sum_{x \le t} p_X(x)$ for *all real* t. It is a nondecreasing, right-continuous step function, 0 left of support and 1 right of support. At x, jump size $P(X = x) = F(x) - F(x^-)$, where $F(x^-) = \lim_{t \uparrow x} F(t)$. Draw a filled dot at the step's left end and an open dot at its right end.

Endpoint dictionary (also works for mixed RVs). $P(a < X \le b) = F(b) - F(a)$; $P(a \le X \le b) = F(b) - F(a^-)$; $P(a < X < b) = F(b^-) - F(a)$; $P(X > b) = 1 - F(b)$; $P(X \ge b) = 1 - F(b^-)$. At a continuous point, $F(b^-) = F(b)$.

9. A maximum from a sample

Choose k distinct numbers uniformly from $1, \dots, n$ without replacement. Total unordered samples $\binom{n}{k}$. If M is the maximum, $M \le m$ means all k draws come from $1, \dots, m$:

$$P(M \le m) = \frac{\binom{m}{k}}{\binom{n}{k}}, \qquad P(M = m) = \frac{\binom{m-1}{k-1}}{\binom{n}{k}}.$$

For real t, use $m = \lfloor t \rfloor$; take $F = 0$ below the smallest feasible maximum and $F = 1$ at/above n. An independent *with-replacement* maximum instead has $P(M \le m) = P(X_1 \le m)^k$.

HW: 2 of $\{1, 2, 3, 4, 5\}$: $\binom{5}{2} = 10$; $p_M(m) = (m-1)/10$ for $m = 2, 3, 4, 5$. CDF values on $[2, 3), [3, 4), [4, 5), [5, \infty)$ are $1/10, 3/10, 6/10, 1$. The PMF bars have heights .1, .2, .3, .4; the CDF jumps by those heights.

Useful counting-to-PMF patterns. n independent yes/no trials, success probability p: $P(K = k) = \binom{n}{k} p^k (1-p)^{n-k}$. Sample r without replacement from N with K marked: $P(X = x) = \binom{K}{x}\binom{N-K}{r-x} / \binom{N}{r}$ (feasible x only). These follow by counting, not by assuming outcomes like totals are equally likely.

10. Continuous densities and CDFs

PDF $f_X(x) \ge 0$ and $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$. Probability is *area*: $P(a < X < b) = \int_a^b f_X(x)\,dx$. A PDF value is a height, not $P(X = x)$; height may exceed 1. For a continuous RV, $P(X = c) = 0$, so endpoint $<$ / $\le$ choices do not change interval probabilities.

Convert PDF ↔ CDF. $F(x) = \int_{-\infty}^{x} f(t)\,dt$; where differentiable, $f(x) = F'(x)$. $P(a < X < b) = F(b) - F(a)$ and $P(X > a) = 1 - F(a)$. A valid CDF is nondecreasing, right-continuous, with limits $F(-\infty) = 0, F(\infty) = 1$. For a continuous model, F has no jumps.

Piecewise recipe. (1) Find c from the area over *every* support interval $= 1$. (2) For each x, integrate from the far left up to x; carry the earlier interval's mass into later pieces. (3) Below support $F = 0$; above it $F = 1$; across a gap F is flat. (4) Check continuity and endpoint values. Sketch f by its heights/shapes, F by accumulated area/slope $F' = f$.

HW: If $f(x) = 1/4$ on $(0, 2) \cup (4, 6)$, then

$$F(x) = \begin{cases} 0, & x \le 0, \\ x/4, & 0 < x < 2, \\ 1/2, & 2 \le x \le 4, \\ (x-2)/4, & 4 < x < 6, \\ 1, & x \ge 6. \end{cases}$$

The gap $(2, 4)$ has zero density and a flat CDF; total area $2(2/4) = 1$.

11. Read probabilities from a CDF

HW: $F(x) = 0$ for $x < -2$; $(x+2)/4$ for $-2 \le x < 2$; 1 for $x \ge 2$. It has no jumps. Therefore $P(-1 < X < 1) = F(1) - F(-1) = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$; $P(X > 0.5) = 1 - F(0.5) = 3/8$. On its support, $f(x) = F'(x) = 1/4$.

12. Integrate over the asked interval

HW: Train deviation density $f(x) = \frac{3}{256}(16 - x^2)$ for $-4 < x < 4$. Antiderivative: $G(x) = \frac{3}{256}(16x - x^3/3)$. "At least 2 min early" means $X \le -2$: $G(-2) - G(-4) = 5/32$. "1 to 3 min late" means $1 \le X \le 3$: $G(3) - G(1) = 35/128$. "Exactly 1 min early" means $X = -1$: probability 0.

Freund's data displays. Relative frequency = class count/n; cumulative relative frequency adds classes through a boundary. A histogram has adjacent bars; an ogive plots cumulative count/percent. These descriptive displays appear at the end of Chapter 3.

13. Normalize, then integrate

HW: $f_Y(y) = cy^2$ on $0 < y < 3$. Since $c\int_0^3 y^2 dy = 9c = 1$, $c = 1/9$. Thus $F_Y(y) = 0$ for $y \le 0$; $y^3/27$ for $0 < y < 3$; 1 for $y \ge 3$. Then $P(Y > 2) = 1 - F_Y(2) = 19/27$.

HW: $F_X(x) = 1 - e^{-2x}$ for $x > 0$, 0 for $x \le 0$. Differentiate the nonconstant branch: $f_X(x) = 2e^{-2x}$ for $x > 0$ (0 elsewhere). Tail $P(X > 1) = 1 - F_X(1) = e^{-2}$.

14. Joint distributions (Freund Ch. 3)

For two discrete RVs, joint PMF $p(x, y) = P(X = x, Y = y) \ge 0$, $\sum_x \sum_y p(x, y) = 1$. For continuous RVs, joint PDF $f(x, y) \ge 0$, $\iint_{\text{support}} f(x, y)\,dx\,dy = 1$; region probability $P((X, Y) \in R) = \iint_R f(x, y)\,dx\,dy$. Respect the actual support/region when setting limits.

Marginalize out the other variable. Discrete: $p_X(x) = \sum_y p(x, y)$, $p_Y(y) = \sum_x p(x, y)$. Continuous: $f_X(x) = \int f(x, y)\,dy$, $f_Y(y) = \int f(x, y)\,dx$, over feasible support. A joint table's row/column sums are the marginals.

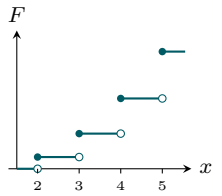
Condition and test independence. $p_{X|Y}(x \mid y) = p(x, y)/p_Y(y)$ when $p_Y(y) > 0$; for densities, $f_{X|Y}(x \mid y) = f(x, y)/f_Y(y)$ when $f_Y(y) > 0$. Independent RVs satisfy $p(x, y) = p_X(x)p_Y(y)$ or $f(x, y) = f_X(x)f_Y(y)$ on the whole support. Check all cells/points and support, not one example point.

Joint CDF. $F_{X,Y}(a, b) = P(X \le a, Y \le b)$: sum joint masses with $x \le a, y \le b$, or integrate f over $(-\infty, a] \times (-\infty, b]$.

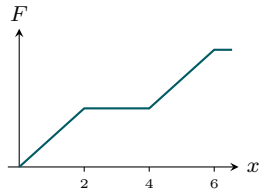
15. Last-minute checks

Discrete: values ≥ 0, masses sum to 1; CDF jumps exactly at possible values.
Continuous: density area $= 1$; CDF continuous and reaches 1; a single point has probability 0.
From CDF: subtract cumulative values for intervals; differentiate only smooth branches, and inspect jumps separately.
Graphs: PMF = bars; discrete CDF = right-continuous stairs; PDF = height/shape; continuous CDF = rising where $f > 0$, flat where $f = 0$.

16. Sketch templates



Discrete maximum CDF



Gap-density CDF

Sources: Freund, 8th ed., Ch. 3; Ross, 8th ed., pp. 168–170 (CDF endpoints).